DBMS Assignment 7

Rollback and Commit commands Create Teacher table with the following fields (Name, DeptNo, Date of joining, DeptName, Location, Salary)

```sql
USE college_db ;
CREATE TABLE Teacher (
    Name VARCHAR(50),
    DeptNo INT,
    DateOfJoining DATE,
    DeptName VARCHAR(50),
    Location VARCHAR(50),
    Salary DECIMAL(10,2)
);
```

(a) Insert five records

```sql
INSERT INTO Teacher VALUES
('Amit', 101, '2020-06-15', 'Mathematics', 'Delhi', 50000),
('Riya', 102, '2019-03-10', 'Commerce', 'Mumbai', 45000),
('Karan', 103, '2021-01-20', 'Mathematics', 'Chennai', 55000),
('Neha', 104, '2018-07-25', 'Commerce', 'Delhi', 40000),
('Rahul', 105, '2022-02-11', 'Science', 'Pune', 48000);
```

(b) Give Increment of 25% salary for Mathematics Department .

```sql
START TRANSACTION;
SET SQL_SAFE_UPDATES = 0;
UPDATE Teacher
SET Salary = Salary + (Salary * 0.25)
WHERE DeptName = 'Mathematics';
```

(c) Perform Rollback command

```sql
ROLLBACK;
```

(d)Give Increment of 15% salary for Commerce Department

```sql
UPDATE Teacher
SET Salary = Salary + (Salary * 0.15)
WHERE DeptName = 'Commerce';
```

(e) Perform commit command

```sql
COMMIT;
```

DBMS Assignment 8

(Exercise on order by and group by clauses) Create Sales table with the following fields (Sales No, Salesname, Branch, Salesamount, DOB)

```sql
CREATE DATABASE SalesDB;
CREATE TABLE Sales (
    SalesNo INT,
    SalesName VARCHAR(50),
    Branch VARCHAR(50),
    SalesAmount DECIMAL(10,2),
    DOB DATE
);
```

(a) Insert five records

```sql
INSERT INTO Sales VALUES
(1, 'Amit', 'Delhi', 5000, '1999-12-21'),
(2, 'Riya', 'Mumbai', 7000, '2000-05-10'),
(3, 'Karan', 'Delhi', 6500, '1998-12-05'),
(4, 'Neha', 'Chennai', 4000, '2001-07-15'),
(5, 'Rahul', 'Mumbai', 8000, '1997-12-30');
```

(b) Calculate total salesamount in each branch

```sql
SELECT Branch, SUM(SalesAmount) AS TotalSales
FROM Sales
GROUP BY Branch;
```

Branch	TotalSales
Delhi	11500.00
Mumbai	15000.00
Chennai	4000.00

(c) Calculate average salesamount in each branch

```sql
SELECT Branch, AVG(SalesAmount) AS AvgSales
FROM Sales
GROUP BY Branch;
```

Branch	AvgSales
Delhi	5750.000000
Mumbai	7500.000000
Chennai	4000.000000

(d) Display all the salesmen, DOB who are born in the month of December as day in character format i.e. 21-Dec-09

```
SELECT SalesName, DATE_FORMAT(DOB, '%d-%b-%y') AS DOB
FROM Sales
WHERE MONTH(DOB) = 12;
```

SalesName	DOB
Amit	21-Dec-99
Karan	05-Dec-98
Rahul	30-Dec-97

(e) Display the name and DOB of salesman in alphabetical order of the month.

```
SELECT SalesName, DOB
FROM Sales
ORDER BY MONTHNAME(DOB);
```

SalesName	DOB
Amit	1999-12-21
Karan	1998-12-05
Rahul	1997-12-30
Neha	2001-07-15
Riya	2000-05-10

DBMS Assignment 9

Consider the following tables namely “DEPARTMENTS” and “EMPLOYEES” Their schemas

are as follows,

Departments (dept no , dept name , dept_location); Employees (emp_id , emp_name , emp_salary,dept_no);

```sql
USE company_db;
CREATE TABLE Departments (
    dept_no INT PRIMARY KEY,
    dept_name VARCHAR(50),
    dept_location VARCHAR(50)
);
CREATE TABLE Employees (
    emp_id INT PRIMARY KEY,
    emp_name VARCHAR(50),
    emp_salary DECIMAL(10,2),
    dept_no INT,
    FOREIGN KEY (dept_no) REFERENCES Departments(dept_no)
);
INSERT INTO Departments VALUES
(1, 'HR', 'Delhi'),
(2, 'Finance', 'Mumbai'),
(3, 'IT', 'Bangalore');
INSERT INTO Employees VALUES
(101, 'Amit', 50000, 1),
(102, 'Riya', 60000, 2),
(103, 'Karan', 55000, 3);
```

a) Develop a query to grant all privileges of employees table into departments table

```sql
GRANT ALL PRIVILEGES
ON Employees
TO 'user1'@'localhost';
```

b) Develop a query to grant some privileges of employees table into departments table

```sql
GRANT SELECT, INSERT
ON Employees
TO 'user1'@'localhost';
```

c) Develop a query to revoke all privileges of employees table from departments table

```
REVOKE ALL PRIVILEGES
ON Employees
FROM 'user1'@'localhost';
```

d) Develop a query to revoke some privileges of employees table from departments table

```
REVOKE UPDATE
ON Employees
FROM 'user1'@'localhost';
```

e) Write a query to implement the save point.

```
SAVEPOINT sp1;

UPDATE Employees
SET emp_salary = emp_salary + 5000
WHERE emp_id = 101;

-- rollback to savepoint
ROLLBACK TO sp1;

COMMIT;
```

Assignment 10

Using the tables “DEPARTMENTS” and “EMPLOYEES” perform the following queries

a) Display the employee details, departments that the departments are same in both the emp and dept.

```
SELECT *
FROM Employees e
INNER JOIN Departments d
ON e.dept_no = d.dept_no;
```

emp_id	emp_name	emp_salary	dept_no	dept_no	dept_name	dept_location
101	Amit	50000.00	1	1	HR	Delhi
104	Amit	50000.00	1	1	HR	Delhi
102	Riya	60000.00	2	2	Finance	Mumbai
103	Karan	55000.00	3	3	IT	Bangalore

b) Display the employee name and Department name by implementing a left outer join.

```
SELECT e.emp_name, d.dept_name
FROM Employees e
LEFT JOIN Departments d
ON e.dept_no = d.dept_no;
```

emp_name	dept_name
Amit	HR
Riya	Finance
Karan	IT
Amit	HR

c) Display the employee name and Department name by implementing a right outer join.

```
SELECT e.emp_name, d.dept_name
FROM Employees e
RIGHT JOIN Departments d
ON e.dept_no = d.dept_no;
```

emp_name	dept_name
Amit	HR
Amit	HR
Riya	Finance
Karan	IT

d) Display the details of those who draw the salary greater than the average salary

```
SELECT *
FROM Employees
WHERE emp_salary > (
    SELECT AVG(emp_salary)
    FROM Employees
);
```

emp_id	emp_name	emp_salary	dept_no
102	Riya	60000.00	2
103	Karan	55000.00	3
NULL	NULL	NULL	NULL